Social Network Analysis of the Facebook Friendship Network

Module: B107 Data Driven Strategic Decision Making

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GitHub link: https://github.com/ADEYEMIjamiu/facebook-network-analysis

1. Introduction

Social networks are vital to comprehending how people connect and share information today, especially in the digital age. Social media sites, including Facebook, Twitter and LinkedIn, provide a wealth of relational data that can be modelled and interpreted using network science techniques. Social network analysis (SNA) is a widely used method to carry out relational structures and also connectivity patterns (Newman, 2010).

In network analysis, users or some other related entities are represented as nodes and the relationships between them are given as edges. By Using graph theory techniques, student and researchers and others can examine structural properties examples like degree distribution, clustering, shortest paths, and centrality measures.

These metrics give insight into how networks develop, how communities are form, and how influence is distributed within social systems. The objective of this project is to examine or analyse a real-life social network dataset using a systematic network analysis process.

The dataset we are using in this study is the Facebook friendship network as made available by Stanford Network Analysis Project (SNAP). Made available in Python and working with the NetworkX library, the project builds the network, performs some statistical analysis and compares it with 3 graph models: (ER), (BA) and (WS) structures realized on network.

However, the research question guiding this analysis is:

Does the Facebook friendship network exhibit small-world and scale-free structural properties when compared with classical random network models?

To answer this question, we will use several network metrics that are computed, including degree distribution, connected components, shortest path statistics, clustering coefficients, and centrality measures. Our findings would then be compared with synthetic networks generated using ER, BA, and WS models to check which theoretical model best explains the structure of the Facebook network.

2. Dataset and Network Construction

Has said the dataset used in this study is the **Facebook Social Circles dataset** from the Stanford Network Analysis Project (SNAP). (Leskovec and McAuley, 2012). The dataset is publicly available at the website or link below:

https://snap.stanford.edu/data/ego-Facebook.html

This dataset will contain anonymised friendship connections among Facebook users and is widely used in network science research. It is provided as an **edge list**, where each row will now represent a friendship connection between two users.

this network representation is:

1. **Nodes** represent Facebook users.
2. **Edges** represent friendship relationships between users.

Also, the dataset does not contain additional node attributes or edge weights; so, the resulting network is treated as an **undirected and unweighted graph**.

After we load the dataset using the NetworkX library in Python, the network appears to contain:

1. **4039 nodes (users)**
2. **88234 edges (friendship connections)**

So therefore, the density of the network is approximately 0.0108, which shows that only a small fraction of all possible connections between users are present. This type is typical for real-world social networks, where users tend to maintain certain connections with a low number of individuals rather than forming connections with every other user.

Meanwhile the network was constructed directly from the edge list using the NetworkX function read_edgelist(). The resulting graph structure represents the underlying social connectivity among Facebook users and forms the basis for more further analysis.

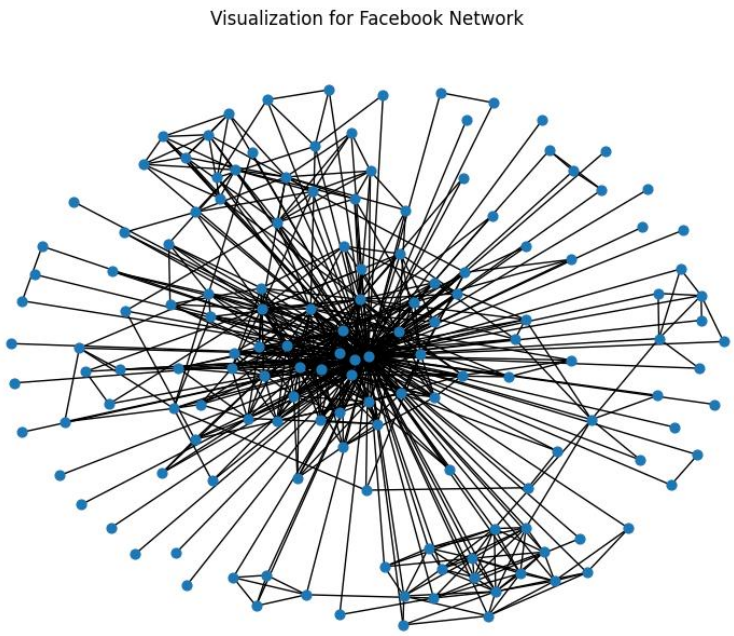

Figure 1: Visualization of the Facebook friendship network

The visualisation shows the complex and highly interconnected structure of the network, with densely connected regions indicating clusters of users.

3. Methodology

The analysis was conducted using **Python in a Google Colab**, by using the **NetworkX** library for graph analysis. Some additional libraries like **NumPy**, **Pandas**, and **Matplotlib** are also used for statistical calculations and some important visualisations.

Some of the analysis pipeline include the following stages:

1. Network construction from the dataset
2. Basic network statistics computation
3. Degree distribution analysis
4. Connected components analysis
5. Shortest path analysis
6. Clustering coefficient and density analysis
7. Centrality analysis
8. Comparison with synthetic graph models

These metrics provide good and deep info into the structure, connectivity, and influence patterns within the network.

For us to interpret path-based metrics such as average shortest path length and network diameter, the analysis was performed on the **(LCC- largest connected component)** of the network. This pattern of approach always ensures that path calculations are meaningful because shortest paths cannot be defined between nodes that are disconnected

lastly, the computed statistics were compared by using these 3 classical network models:

1. **(ER) random graph model**
2. **(BA) scale-free model**
3. **(WS) small-world model**

These 3 models are generated with approximately the same number of nodes & edges as the real Facebook network to provide a useful structural comparison.

We performed these computations by using NetworkX functions, by ensuring reproducibility and further consistency in our analysis.

So we can say this structured approach allows for a systematic comparison between empirical and theoretical network models.

4. Network Analysis Results

4.1 Basic Network Properties

As said and showed on the uploaded dataset the Facebook network contains **4039 nodes and 88234 edges**, with a density of approx. **0.0108**. The relatively low density shows that the network is sparse, which are common characteristic of social networks where individuals maintain a limited numbers of connections relative to the total population.

Despite this sparsity, the network forms just a **single connected component**, its means that every user can reach every other user through chains of friendships. The presence of a giant connected component is however always in large social networks and indicates a very high level of global connectivity.

4.2 Degree Distribution

So , Degree distribution describes how many connections each node has within the network. In this Facebook network, the average degree is approx. **43.69**, while the maximum observed degree exceeds **1000 connections**. So this pattern considers to be characteristic of scale-free networks (Barabási and Albert, 1999).

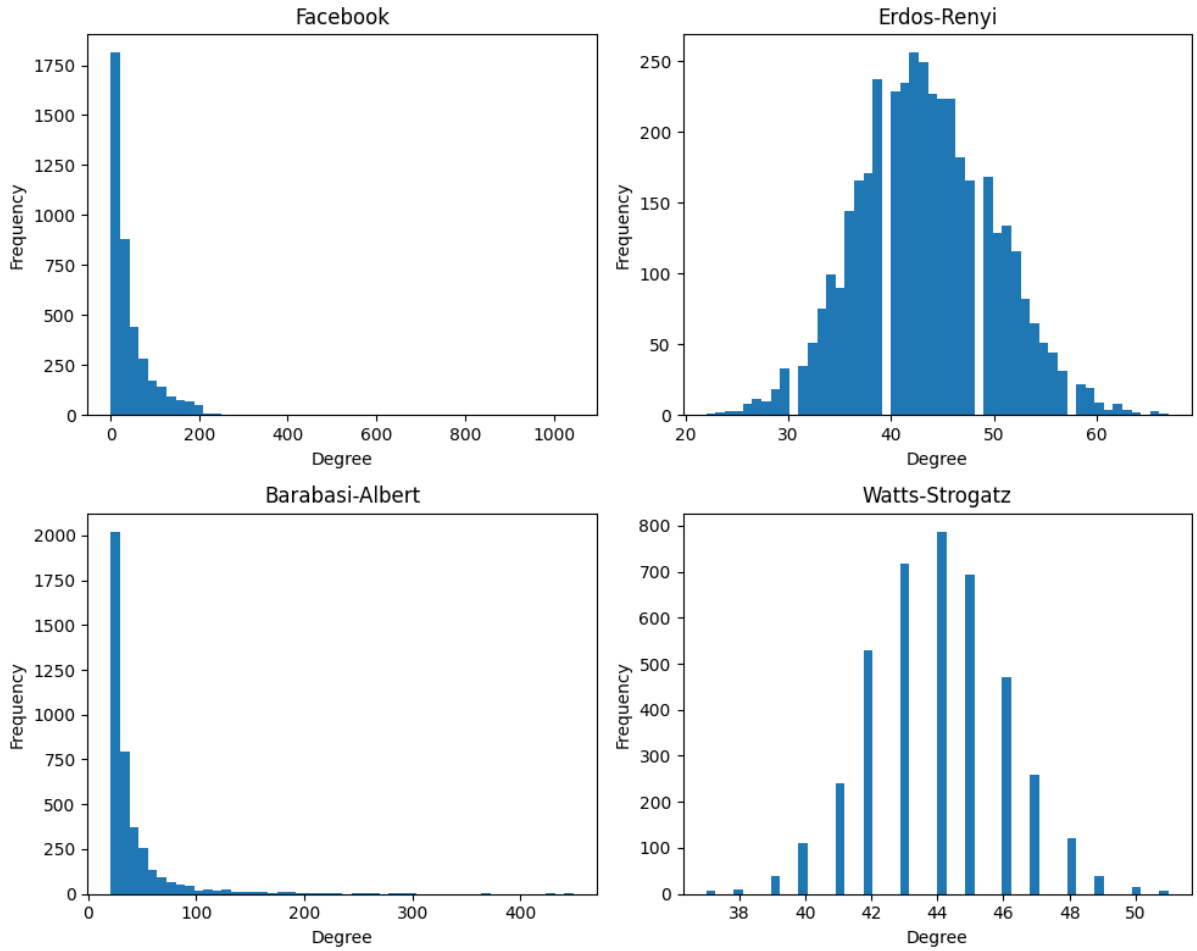

Figure 2: Degree distribution of the Facebook friendship network

This image of histogram illustrates how the number of connections varies across users in the network. As shown here in Figure 2, the distribution is highly skewed.

Source: Author's own analysis using the SNAP Facebook dataset.

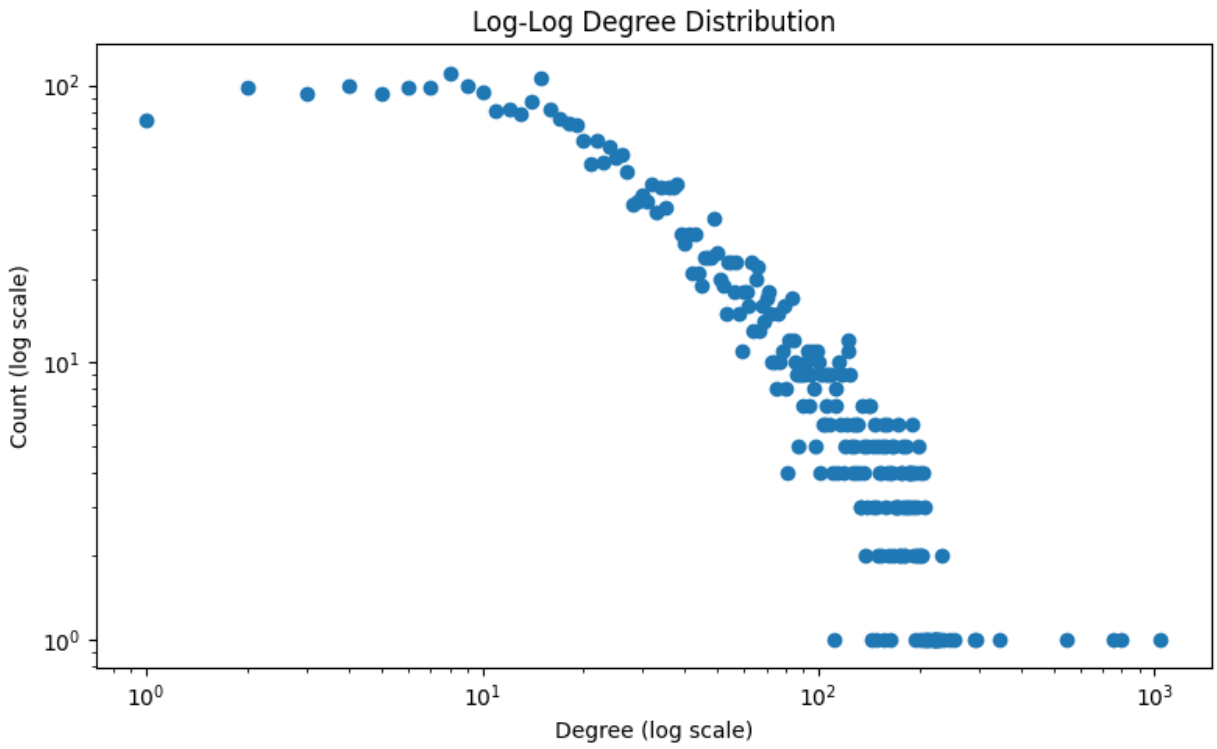

Figure 3: Log-log degree distribution of the Facebook network

Figure 3 further confirms the heavy-tailed nature of the distribution.

The histogram of the degree distribution shows a strongly skewed pattern. Showing that Most users have relatively few connections, while a less or low number of users possess extremely high numbers of connections. When visualised on a log-log scale, the distribution shows and demonstrate a heavy-tailed pattern.

This pattern is part of **scale-free networks**, where a small number of highly connected nodes (that we often refer as hubs) dominate the network structure. These hubs play a significant role to make sure it maintains the connectivity and stability of the network.

4.3 Path Analysis

Shortest path explain or analyse measures the minimum number of steps required to travel between nodes in the network.

The Facebook network exhibits:

1. **Average shortest path length:** is approximately **3.69**
2. **Network diameter: 8**

These results shows and indicate that most users can reach each other within only a few friendship connections. This phenomenon is often referred to as the **"small-world effect"**, where even large networks surprisingly and maintain short path lengths between nodes.

The small-world property suggests that information or influence can spread rapidly across the network really quickly

4.4 Clustering Coefficient and Density

Here we can see that the average clustering coefficient of the network is approximately **0.605**, indicating a high level of local clustering.

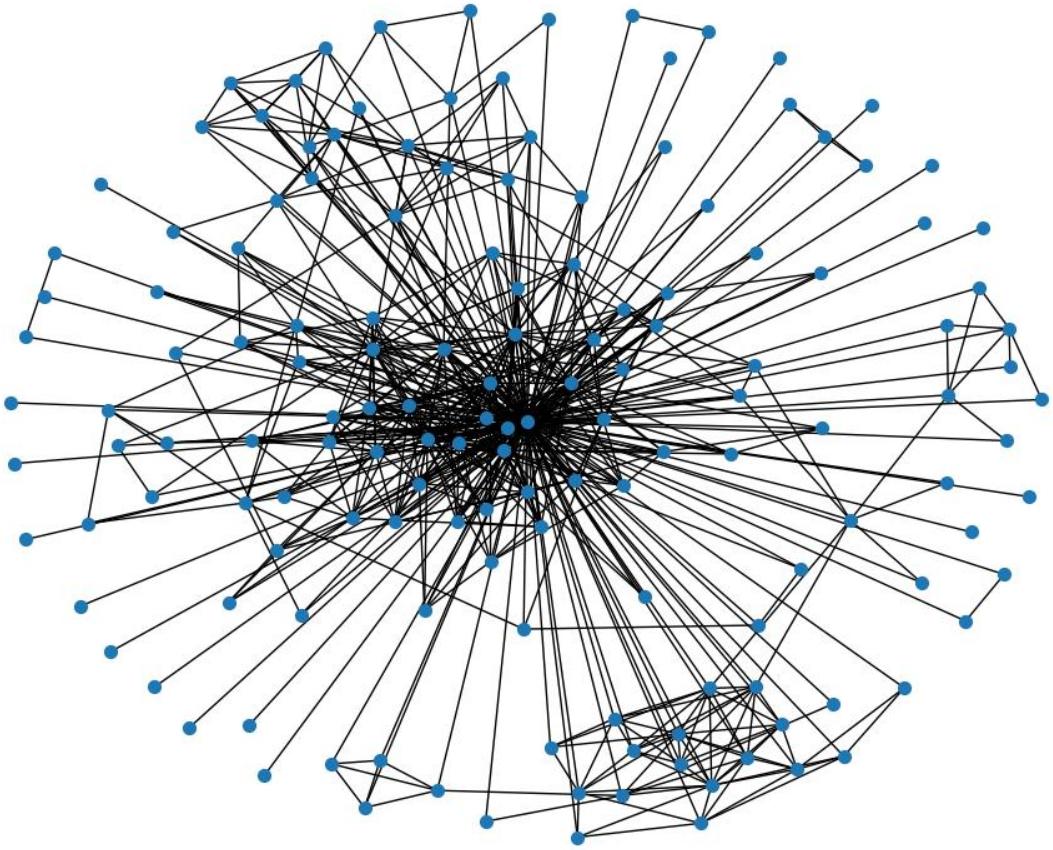

Figure 4: Image visualization of clustered communities within the Facebook network

Figure 4 image illustrates the presence of tightly connected communities within the network. These clusters reflect groups of users with strong mutual connections, supporting the high clustering coefficient observed in the analysis.

Source: Author's own analysis using the SNAP Facebook dataset.

Clustering measures the likelihood that two friends of a given user are also friends with each other. In social networks, high clustering shows the tendency for individuals to create tightly connected communities or social circles.

Even though that the overall network density is low, the high clustering coefficient still shows that local neighbourhoods within the network are densely connected. This combination of low global density and high local clustering is a defining characteristic of many real-world social networks.

4.5 Centrality Analysis

Centrality measures identify the most influential or structurally important nodes in a network. Several centrality metrics are being calculated, and its includes:

1. **Degree centrality**
2. **Closeness centrality**
3. **Betweenness centrality**

Nodes with more higher degree centrality represent users with many connections. These individuals act as hubs within the network.

Some Nodes with high betweenness centrality serve as bridges between different regions f network. These nodes are consider very important because they speedily facilitate the flow of information between otherwise disconnected groups.

The report analysis identified several highly central nodes that likely function as structural connectors linking different friendship communities.

5. Comparison with Network Models

For us to better understand the structure of the Facebook network, its statistics were compared with 3 classical graph models.

Table 1: provides a comparative overview of key structural metrics across the Facebook network and the synthetic models.

Graph	Nodes	Edges	Density	Avg Clustering	Avg Path Length	Diameter	Avg Degree
FB	4039	88234	0.0108	0.6055	3.69	8	43.69
ER	4039	87553	0.0107	0.0107	2.61	4	43.35
BA	4039	88374	0.0108	0.0382	2.50	4	43.76
WS	4039	88858	0.0109	0.5347	2.96	4	44.00

	Graph	Nodes	Edges	Density	Average Clustering	Connected Components	Average Path Length (LCC)	Diameter (LCC)	Average Degree
0	FB	4039	88234	0.010820	0.605547	1	3.692507	8	43.691013
1	ER	4039	87553	0.010736	0.010766	1	2.610505	4	43.353800
2	BA	4039	88374	0.010837	0.038214	1	2.507594	4	43.760337
3	WS	4039	88858	0.010896	0.534717	1	2.964702	4	44.000000

Erdős–Rényi Model

Here, The ER model represents purely random networks where edges are formed with equal probability. Compared with the Facebook network, the ER graph exhibits extremely low clustering and a more uniform degree distribution. This demonstrates a fundamental limitation of random graph models in capturing the structured and community-driven nature of real-world social networks.

Barabási–Albert Model

In this case the BA model generates scale-free networks through a process known as preferential attachment. In this model, new nodes tend to connect to already highly connected nodes, producing hub-like structures.

The Facebook network shares similarities with the BA model in its heavy-tailed degree distribution and the presence of highly connected nodes.

Watts–Strogatz Model

Furthermore, the WS model is designed to reproduce the **small-world phenomenon**, Joining high clustering with short path lengths. The clustering coefficient and path length observed in the Facebook network closely resemble those produced by the WS model. This phenomenon is consistent with the small-world property described by Watts and Strogatz (1998).

The coexistence of high clustering and hub dominance suggests that real-world social networks cannot be fully explained by a single theoretical model but instead exhibit hybrid structural properties.

Lastly, this highlights that real-world networks are shaped by both global structural constraints and local interaction patterns, which are not fully captured by simplified theoretical models.

6. Research Question

So the research question from the starts of this project was:

Does the Facebook friendship network exhibit small-world and scale-free properties when compared with classical network models?

According to the results, its strongly suggest that the Facebook network exhibits characteristics of both network types.

The heavy-tailed degree distribution indicates the presence of highly connected hub nodes, consistent with **scale-free networks**.

Also At the same time, the network demonstrates **short average path lengths and high clustering**, which are defining features of **small-world networks**.

So therefore, the Facebook network appears to combine structural properties from both the BA and WS models.

7. In Conclusion

Finally, we say this project analysed the structure of the Facebook friendship network using a systematic social network analysis pipeline implemented in Python. The dataset consisted of over four thousand nodes and more than eighty thousand friendship connections.

The analysis revealed several key structural properties. The network forms a single connected component, indicating strong overall connectivity. The degree distribution is highly skewed, suggesting the presence of influential hub nodes. Moreso the network also exhibits short average path lengths and high clustering, reflecting the small-world phenomenon observed in many real-world social networks.

When compared with classical network models, the Facebook network differs significantly from purely random graphs but shows similarities with both scale-free and small-world models. These findings suggest that social networks evolve through a combination of preferential attachment and community-based clustering.

So Overall, the findings demonstrate that the Facebook network exhibits complex structural characteristics that cannot be captured by a single theoretical model. Instead, it combines features of both scale-free and small-world networks, reflecting the interplay between preferential attachment and local clustering in real-world social systems.

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